Notes: Aghion and Bolton (RES, 1992)

An Incomplete Contracts Approach to Financial Contracting

Contents

1 Big picture **3**

2 Data and measurement (key objects) **4**

2.1 This is a theory paper: what the key objects are 4
2.2 Agents and preferences 4
2.3 States, actions, signal, and returns 5
2.4 First-best benchmark 5
2.5 Contractual objects 6
2.6 Timing 6

3 Models and empirical specifications (all equations + interpretation) **7**

3.1 Basic environment and first-best action choice 7
3.2 Contractual incompleteness and the role of the public signal 7
3.3 Why renegotiation makes control rights central 8
3.4 Entrepreneur control 8
3.5 Investor control 9
3.6 Contingent control and the logic of debt 10
3.7 Why joint ownership is dominated in the baseline model 11
3.8 Verifiable actions and ex ante action restrictions 12
3.9 Financial interpretation of the model 12

4 Identification and instruments (what makes the estimates credible) **13**

4.1 This is a theory paper: what pins down the results 13
4.2 Why wealth constraints are essential 13
4.3 Why renegotiation is essential 13
4.4 Why the signal matters 14

4.5 Why comonotonicity assumptions are useful benchmarks 14
4.6 Why joint ownership fails in the baseline 14
4.7 Why message-contingent “Maskin” mechanisms do not solve the problem 14
4.8 Limits of the model and why they matter 14

5 Tables and figures **15**
5.1 Figure 1: Timeline of the contracting problem 15
5.2 Numerical example in Section III.A 15
5.3 Investor-control threshold in the example 16
5.4 The proposition map: when each governance structure is optimal 16
5.5 Financial-contract interpretation 16

6 Short summary **16**

7 Conclusion **17**
7.1 Core conclusion (what the paper establishes) 17
7.2 Economic intuition 17
7.3 Takeaway 17

1 Big picture

- **Question.** How should a firm allocate **control rights** between an entrepreneur and outside investors when financial contracts are **incomplete**, the entrepreneur has **private benefits of control**, and she has **no initial wealth**? Why do familiar instruments such as **debt**, **voting equity**, and **non-voting equity** emerge as sensible financing choices?

- **Setting.** The paper studies a bilateral contract between a **penniless entrepreneur** and a **wealthy investor**. The entrepreneur needs outside funding to cover the fixed set-up cost K. After financing is raised, a state of nature is realized and a future action must be chosen. The contract cannot be written directly on the state, but it can be written on a **publicly verifiable signal** that is correlated with the state.

- **Core challenge.** Cash-flow rights alone do not solve the problem. The entrepreneur cares about both monetary payoffs and **non-pecuniary private benefits**, while the investor cares only about monetary returns. Because the contract is incomplete and the entrepreneur is wealth constrained, the parties cannot always write a transfer scheme that perfectly aligns incentives.

- **Main theoretical contribution.** The paper builds a **control-rights theory of capital structure**. It shows that the central problem in financial contracting is not only how to split cash flows, but also how to allocate the right to decide future actions that were not fully specified ex ante.

- **Main mechanism.** Control matters because it determines the **status quo point in renegotiation**. Since the entrepreneur has the bargaining power in renegotiation, a control allocation that looks good ex post may still fail ex ante if it leaves the investor with too little pledgeable income.

- **Why it matters.** This paper is one of the foundational treatments of the idea that **capital structure is a governance structure**. Debt is valuable not just because it promises a fixed repayment, but because **default transfers control**. Equity matters not just because it shares upside, but because different forms of equity carry different **voting rights**.

- **Main results.**
  - If the entrepreneur's **private benefits are aligned** with total surplus, full **entrepreneur control** is feasible and efficient.
  - If **monetary returns are aligned** with total surplus, full **investor control** can implement the first best.
  - When neither alignment condition holds, **entrepreneur control may not be feasible** and **investor control may be too rigid** because the entrepreneur cannot afford to compensate the investor in renegotiation.
  - In that case, **contingent control** – giving control to the entrepreneur when the signal is good and to the investor when the signal is bad – can dominate both unilateral allocations.
  - With only two actions, ex ante **action restrictions** are redundant. But with richer action sets, **covenants and restrictions** can improve efficiency by shrinking renegotiation rents.

- **Financial interpretation.** The paper maps governance structures into securities:
  - entrepreneur control $\leftrightarrow$ **non-voting equity / preferred stock**,
  - investor control $\leftrightarrow$ **voting equity**,

- contingent control $\leftrightarrow$ **debt**, **convertible debt**, **warrants**, or **convertible preferred stock**,
- action restrictions $\leftrightarrow$ **debt covenants and charter restrictions**.

- **Economic takeaway.** The optimal financial contract is the one that best trades off **ex post efficiency** against **ex ante pledgeability**. Control rights are valuable because they limit future rent extraction and therefore determine how much outside finance can be raised.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no empirical dataset**. The paper is a pure theory paper.
- The central objects are:
  - the entrepreneur and investor,
  - the state of nature θ,
  - the future action a,
  - the verifiable signal s,
  - monetary returns r,
  - the entrepreneur's private benefits $l(a, \theta)$,
  - transfer rules t, and
  - control-right allocations.
- The whole paper is about how these primitives determine the feasible set of **governance structures** and, through them, the feasible set of **financial contracts**.

2.2 Agents and preferences

- **Entrepreneur E.**
  - Has a project but no initial wealth.
  - Needs outside funding K from an investor.
  - Cares about both money and private benefits from how the project is run.
- **Investor I.**
  - Is wealthy and can finance the project.
  - Cares only about monetary returns.
- Both are risk neutral. Utility functions are:

$$U_E(r, a) = r + l(a, \theta), \qquad U_I(r, a) = r.$$

2.3 States, actions, signal, and returns

- There are **two states of nature**:
$$\Theta = \{\theta_g, \theta_b\}.$$
- There are **two actions**:
$$A = \{a_g, a_b\},$$
where a_g is the first-best action in state θ_g and a_b is the first-best action in state θ_b.
- There is a **binary signal** realized at date 1:
$$s \in \{0, 1\}.$$
It is publicly verifiable and correlated with the state:
$$\beta^g \equiv \mathbb{P}(s = 1 \mid \theta = \theta_g) > \frac{1}{2}, \qquad \beta^b \equiv \mathbb{P}(s = 1 \mid \theta = \theta_b) < \frac{1}{2}.$$
- There is a **binary monetary return** at date 2:
$$r \in \{0, 1\}.$$
- The expected monetary return in state θ_i under action a_j is
$$y^i_j \equiv \mathbb{E}[r \mid \theta = \theta_i, a = a_j] = \mathbb{P}(r = 1 \mid \theta = \theta_i, a = a_j).$$
- The entrepreneur's private benefit in state θ_i under action a_j is denoted by
$$l^i_j.$$

2.4 First-best benchmark

- In state θ_g, the total-surplus maximizing action is a_g:
$$y^g_g + l^g_g > y^g_b + l^g_b.$$
- In state θ_b, the total-surplus maximizing action is a_b:
$$y^b_b + l^b_b > y^b_g + l^b_g.$$
- Let $q \equiv \mathbb{P}(\theta = \theta_g)$. The project is first-best feasible if expected first-best monetary returns are large enough to cover the initial set-up cost:
$$q y^g_g + (1 - q) y^b_b > K.$$

2.5 Contractual objects

- **Transfer schedule.**
  - With unverifiable actions, the entrepreneur receives $t(s, r) \geq 0$.
  - Because $r \in \{0, 1\}$, there is no loss of generality in writing

$$t(s, r) = t_s r + t'_s.$$

  - The nonnegativity restriction reflects the entrepreneur's **zero wealth**: she cannot make negative transfers.
- **Control allocation.**
  - After signal realization s, either the entrepreneur or the investor may get the right to choose the action.
  - Let $\alpha_s \in [0, 1]$ denote the probability that the entrepreneur controls when signal s is realized.
  - Then $1 - \alpha_s$ is the investor's control probability.
- **Joint control.**
  - The paper also allows both parties to have simultaneous control rights.
  - In that case, action choice requires unanimous consent; disagreement leads to a standstill with payoff $(0, 0)$.
- **Renegotiation.**
  - After the true state is learned, the entrepreneur makes a take-it-or-leave-it offer to the investor.
  - This means the entrepreneur captures all gains from renegotiation, subject to the investor's participation constraint.

2.6 Timing

- The basic timeline is:
  1. at $t = 0$, investment K is made;
  2. at $t = 1$, the state θ and signal s are realized;
  3. after the signal, an action a is chosen;
  4. at $t = 2$, monetary return r is realized.
- This timing is crucial because the contract is written before the state is known, but the actual action choice occurs after new information arrives.

3 Models and empirical specifications (all equations + interpretation)

3.1 Basic environment and first-best action choice

The entrepreneur's and investor's utilities are

$$U_E(r, a) = r + l(a, \theta), \qquad U_I(r, a) = r.$$

In each state the first-best action maximizes total surplus:

$$a^*(\theta) = \arg\max_{a \in A} \{\mathbb{E}[r \mid a, \theta] + l(a, \theta)\}.$$

With the two-state two-action notation, this gives:

$$y_g^g + l_g^g > y_b^g + l_b^g, \qquad y_b^b + l_b^b > y_g^b + l_g^b.$$

Interpretation.

- The paper distinguishes sharply between **monetary returns** and **total surplus**.
- The investor cares only about y_j^i, while efficiency depends on $y_j^i + l_j^i$.
- This gap is exactly why control rights matter: the action preferred by the cash-flow claimant need not be the action that maximizes total value.

3.2 Contractual incompleteness and the role of the public signal

The contract cannot be made contingent on the realized state θ directly. Instead, it can be made contingent on a verifiable signal $s \in \{0, 1\}$ with

$$\beta^g \equiv \mathbb{P}(s = 1 \mid \theta_g) > \frac{1}{2}, \qquad \beta^b \equiv \mathbb{P}(s = 1 \mid \theta_b) < \frac{1}{2}.$$

The paper notes that the degree of incompleteness can be summarized by the distance

$$d\big((\beta^g, \beta^b), (1, 0)\big) = |1 - \beta^g| + |0 - \beta^b|.$$

Interpretation.

- If $(\beta^g, \beta^b) = (1, 0)$, the signal perfectly reveals the state and the contract is effectively complete.
- If $\beta^g = \beta^b$, the signal is uninformative and contingent control becomes useless.
- In between, the contract can only use a noisy proxy for the true state, which is why contingent control may help but cannot always achieve the first best exactly.

3.3 Why renegotiation makes control rights central

Suppose the contract specifies a transfer schedule and a control allocation. Once the state θ is observed, the parties may renegotiate. Because the entrepreneur makes a take-it-or-leave-it offer, the investor only gets her **status quo payoff**, while the entrepreneur captures the gains from moving to a better action.

Interpretation.

- Control matters not just because it determines who picks the action in the absence of renegotiation.
- More deeply, control determines the **disagreement point** and therefore the division of surplus in renegotiation.
- This is the key insight of the paper: capital structure matters because it changes bargaining positions after uncertainty is resolved.

3.4 Entrepreneur control

Under entrepreneur control, ex post renegotiation always pushes the action toward efficiency whenever the entrepreneur initially chooses the wrong action, because the investor can be held to her status quo payoff and the entrepreneur captures the extra surplus.

3.4.1 Aligned private benefits: Proposition 1

If private benefits are comonotonic with total returns,

$$l_g^g > l_b^g, \qquad l_b^b > l_g^b,$$

then entrepreneur control is always feasible. The entrepreneur can simply be paid a constant transfer t independent of s and r.

Interpretation.

- If the entrepreneur's non-pecuniary preferences are aligned with efficiency, then giving her control causes no distortion.
- Renegotiation is not needed in equilibrium, and the investor's participation constraint can be satisfied by choosing the constant transfer appropriately.

3.4.2 Misaligned private benefits: Proposition 2

Suppose, without loss of generality, that

$$l_b^b < l_g^b,$$

so that private benefits are *not* comonotonic with total returns in state θ_b.

Define

$$\Delta^b \equiv (y_b^b + l_b^b) - (y_g^b + l_g^b), \qquad \Delta_y^b \equiv y_b^b - y_g^b.$$

The paper shows that the investor's best expected payoff under entrepreneur control is the maximum of three candidate values:

$$\pi_1 = \left[q y_g^g + (1-q) y_b^b\right] \frac{\Delta^b}{\Delta_y^b},$$

$$\pi_2 = q y_g^g + (1-q) y_g^b,$$

$$\pi_3 = q \left[\beta^g y_g^g + (1-\beta^g) y_g^g \frac{\Delta^b}{\Delta_y^b}\right] + (1-q) \left[\beta^b y_g^b + (1-\beta^b) y_b^b \frac{\Delta^b}{\Delta_y^b}\right].$$

Then:

$$\text{Entrepreneur control is feasible if and only if} \qquad \max\{\pi_1, \pi_2, \pi_3\} \geq K.$$

Interpretation.

- π_1 corresponds to a **renegotiation-proof** contract.
- π_2 corresponds to a contract with **full renegotiation** in state θ_b.
- π_3 corresponds to a contract with **partial renegotiation** in state θ_b.

The surprising result is that a renegotiation-proof contract can be **strictly dominated** by a contract that allows some renegotiation, because preventing renegotiation requires handing too much of the project's cash flow to the entrepreneur.

Economic meaning.

- Entrepreneur control is best for ex post flexibility, but it may be bad for ex ante finance.
- The reason is simple: if the entrepreneur keeps control and also captures renegotiation gains, the investor may not be able to recover the initial investment K.

3.5 Investor control

Investor control is the opposite polar case. It protects the investor against opportunistic actions by the entrepreneur, but it may be too rigid because the entrepreneur may be too poor to buy back efficient decisions in renegotiation.

3.5.1 Aligned monetary returns: Proposition 3

If monetary returns are comonotonic with total returns,

$$y_g^g > y_b^g, \qquad y_b^b > y_g^b,$$

then investor control achieves the first best.

Interpretation.

- If the investor's preferred action is already the efficient one in every state, giving the investor control creates no inefficiency.
- In this case governance can be made entirely investor-oriented without sacrificing total surplus.

3.5.2 Misaligned monetary returns: Proposition 4

Now suppose monetary returns are *not* comonotonic with total returns, and without loss of generality

$$y_g^g < y_b^g.$$

To induce the investor to switch from the status-quo action a_b to the first-best action a_g in state θ_g, the entrepreneur must offer a lower transfer $\hat{t}_s$ satisfying

$$(1-\hat{t}_s)y_g^g \geq (1-t_s)y_b^g,$$

so that

$$\hat{t}_s \leq 1-(1-t_s)\left(\frac{y_b^g}{y_g^g}\right). \tag{5}$$

Because the entrepreneur cannot make negative transfers, renegotiation is feasible only if

$$t_s \geq 1-\frac{y_g^g}{y_b^g}.$$

This yields the investor-control feasibility condition:

$$\pi_4 \equiv \left(qy_b^g+(1-q)y_b^b\right)\frac{y_g^g}{y_b^g} \geq K.$$

Interpretation.

- Investor control may fail not because the efficient action is unknown, but because the entrepreneur is too poor to compensate the investor for choosing it.
- In other words, investor control can generate ex post inefficiency **because of the entrepreneur's wealth constraint**.

3.6 Contingent control and the logic of debt

The paper's most distinctive result is that a **contingent control allocation** can dominate both unilateral control allocations.

Consider the control rule

$$\alpha_1 = 1, \qquad \alpha_0 = 0,$$

that is, the entrepreneur controls when $s=1$ and the investor controls when $s=0$. Suppose also that

$$t(s,r)=0 \qquad \text{for all } s,r.$$

Then the induced action plan in the absence of renegotiation is:

$$\text{in state } \theta_g, \qquad a=\begin{cases} a_g & \text{if } s=1, \\ a_b & \text{if } s=0, \end{cases}$$

$$\text{in state } \theta_b, \qquad a=\begin{cases} a_g & \text{if } s=1, \\ a_b & \text{if } s=0. \end{cases}$$

The investor's expected return from this action plan is

$$\pi_c = q\big[\beta^g y_g^g + (1-\beta^g) y_b^g\big] + (1-q)\big[\beta^b y_g^b + (1-\beta^b) y_b^b\big]. \tag{6}$$

When the signal becomes highly informative, $(\beta^g, \beta^b) \to (1, 0)$, we have

$$\pi_c \to q y_g^g + (1-q) y_b^b,$$

that is, the first-best expected monetary return.

The paper also shows

$$\pi_c - \pi_3 = q(1-\beta^g)\left[y_b^g - y_g^g \frac{\Delta^b}{\Delta_y^b}\right] + (1-q)(1-\beta^b)\left[1 - \frac{\Delta^b}{\Delta_y^b}\right] y_b^b > 0,$$

under the relevant parameter configuration.

Proposition 5. When neither monetary nor private returns are comonotonic with total benefits, there exist values of K such that:

1. entrepreneur control is not feasible,
2. investor control is not first-best efficient,
3. and both are dominated by the contingent control allocation $(\alpha_0 = 0, \alpha_1 = 1)$ when $(\beta^g, \beta^b) \to (1, 0)$.

Interpretation.

- Contingent control is valuable because it conditions authority on performance-related information.
- It gives the entrepreneur enough room to enjoy private benefits when prospects are good, while protecting the investor when prospects are bad.
- This is the core reason the paper interprets **ordinary debt** as a useful financial contract: default transfers control.

3.7 Why joint ownership is dominated in the baseline model

In the paper's baseline setup, joint ownership is always weakly dominated by unilateral or contingent control.

Interpretation.

- Under joint control, either party can threaten to veto the action and force the firm to a standstill.
- This inflates ex post bargaining rents.
- Because the entrepreneur has the bargaining power in renegotiation, those rents largely go to the entrepreneur, which makes it difficult or impossible to satisfy the investor's participation constraint.

3.8 Verifiable actions and ex ante action restrictions

When actions are verifiable, the contract may specify transfers contingent on actions,

$$t(a, s, r) = t_s^a r + t_s'^a, \qquad t(a, s, r) \geq 0.$$

The paper then asks whether **ex ante action restrictions** can substitute for control rights.

Proposition 6. When the action set is just

$$A = \{a_g, a_b\},$$

any investment contract with some ex ante action restriction is weakly dominated by either a unilateral or a contingent control contract **without** action restrictions.

The reason is that contingent control generates a more flexible status-quo action plan. In the key case studied in the paper, contingent control induces

$$a_s(\theta_g) = \begin{cases} a_g & \text{if } s = 1, \\ a_b & \text{if } s = 0, \end{cases} \qquad a_s(\theta_b) = a_b \quad \text{for all } s, \tag{7}$$

whereas a pre-determined action plan induces

$$a_s(\theta_g) = a_s(\theta_b) = \begin{cases} a_g & \text{if } s = 1, \\ a_b & \text{if } s = 0. \end{cases} \tag{8}$$

Interpretation.

- With only two actions, formal action restrictions do not buy anything that a clever control rule cannot already achieve.
- But with **more than two actions**, action restrictions can matter a lot. The paper gives an example with a third Pareto-dominated action that yields high private benefits to the entrepreneur; banning that action lowers renegotiation rents and improves financeability.
- This is why the paper ends up rationalizing **debt contracts with covenants**: control rights and action restrictions are often complements.

3.9 Financial interpretation of the model

The paper explicitly maps governance structures into familiar securities:

- **Full investor control** corresponds naturally to **voting equity**.
- **Full entrepreneur control** corresponds naturally to **non-voting equity** or **preferred stock**.
- **Joint ownership** corresponds to arrangements like **partnerships** or **trusts**.
- **Contingent control** corresponds to **ordinary debt**, and more generally to **convertible debt**, **warrants**, and **convertible preferred stock**.

The debt interpretation is especially important. If low first-period outcomes mean bad prospects, then the entrepreneur retains control when she can meet debt obligations and loses control when she defaults. In that sense, the contract reallocates decision rights exactly when the investor most wants protection.

Interpretation.

- The value of debt in this paper is not primarily tax, signalling, or monitoring.
- Its value comes from the **governance structure it induces**. Default is a mechanism for transferring authority.
- The paper also explains why convertibles can be useful: if a firm grows large, investors may need more control than when the firm is small.

4 Identification and instruments (what makes the estimates credible)

4.1 This is a theory paper: what pins down the results

- There is no empirical identification problem here. The relevant issue is what **analytically pins down** the optimal governance structure.
- The paper's logic is disciplined by four ingredients:
  1. **contractual incompleteness**: the contract cannot condition on the true state θ;
  2. **wealth constraints**: the entrepreneur cannot make negative transfers;
  3. **renegotiation**: ex post efficient changes are possible, but bargaining power matters;
  4. **signal informativeness**: contingent control only works if the signal contains useful information.

4.2 Why wealth constraints are essential

- If the entrepreneur were sufficiently wealthy, she could always compensate the investor in renegotiation and buy back the efficient action.
- In that case, the initial control allocation would matter much less, because the first best could be restored by side payments.
- The reason control matters in this paper is precisely that the entrepreneur cannot always afford those payments.

4.3 Why renegotiation is essential

- Without renegotiation, the problem would be a simpler mechanism-design problem about implementing actions.
- With renegotiation, the contract must also account for who captures the gains from changing the status quo after the state becomes known.

- This is why the paper repeatedly emphasizes that the investor's ex ante payoff depends on the **status quo value under the initial control allocation**, not just on the final ex post action.

4.4 Why the signal matters

- The signal s is the only contractible proxy for the uncontractible state θ.
- If s is highly informative, contingent control can mimic state-contingent control quite closely.
- If s is uninformative, contingent control collapses toward a blunt rule and loses its advantage.

4.5 Why comonotonicity assumptions are useful benchmarks

- The two clean benchmark cases are:
  - private benefits aligned with total returns,
  - monetary returns aligned with total returns.
- These cases separate the source of the contracting problem:
  - when private benefits are aligned, entrepreneur control is safe;
  - when money is aligned, investor control is safe;
  - when neither is aligned, contingent control becomes valuable.

4.6 Why joint ownership fails in the baseline

- Joint ownership is not inefficient by assumption. It becomes inefficient because veto power raises hold-up rents.
- Since those rents go to the entrepreneur in renegotiation, joint ownership weakens pledgeability too much in the asymmetric setting studied here.

4.7 Why message-contingent "Maskin" mechanisms do not solve the problem

- In the appendix, the paper argues that even if one allows contracts contingent on the parties' announcements of the state, first-best implementation generally still fails.
- The reason is again the combination of **wealth constraints** and **ex post renegotiation**.
- So the paper is not merely saying "the contract is incomplete because the author chose it to be." It argues that standard implementation tricks do not in general remove the governance problem.

4.8 Limits of the model and why they matter

- The paper deliberately uses a **two-state, two-action** environment so that the optimal governance regions can be fully characterized.
- This simplicity is helpful for clarity, but it rules out more complex arrangements like multiple simultaneous control rights over different decisions.

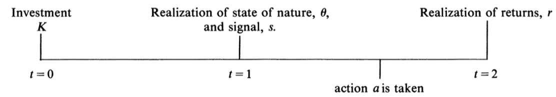

FIGURE 1

- The authors explicitly note that richer action spaces generate more realistic structures such as **debt covenants**, selective veto rights, and more complicated combinations of claims.

5 Tables and figures

5.1 Figure 1: Timeline of the contracting problem

- Figure 1 is extremely simple but conceptually central.
- It lays out the entire sequence:

$$t = 0 : \text{investment } K, \qquad t = 1 : \text{state } \theta \text{ and signal } s \text{ realized, action chosen}, \qquad t = 2 : \text{return } r \text{ realized.}$$

- Read the figure as the whole logic of the paper in one line: finance is committed before the state is known, but control becomes valuable after information arrives.

5.2 Numerical example in Section III.A

- The paper gives a numerical example with

$$q = \frac{1}{2},$$

and payoff parameters:

$$\text{in state } \theta_g : \quad y_g^g = 100,\ l_g^g = 150,\ y_b^g = 200,\ l_b^g = 0,$$

$$\text{in state } \theta_b : \quad y_b^b = 50,\ l_b^b = 0,\ y_g^b = 0,\ l_g^b = 49.$$

- This example shows clearly that private benefits are not comonotonic with total revenues.
- The three candidate investor payoffs under entrepreneur control become:

$$\pi_1 = \frac{3}{2}, \qquad \pi_2 = 50, \qquad \pi_3 = 50\beta^g + 1 - \beta^g + \frac{1 - \beta^b}{2}.$$

- The important lesson is not the exact numbers, but the ranking:
  - renegotiation-proof entrepreneur control can be very bad for the investor,
  - some renegotiation can improve the investor's expected payoff,
  - maximum renegotiation need not be optimal because the entrepreneur captures all bargaining rents.

5.3 Investor-control threshold in the example

- In the same numerical example, the first-best action plan cannot be implemented under investor control whenever
$$K > \left(\frac{1}{2} \cdot 200 + \frac{1}{2} \cdot 50\right) \cdot \frac{1}{2} = 62.5.$$
- This is the clean quantitative illustration of the paper's main point about wealth constraints: even when renegotiation would raise total surplus, the entrepreneur may simply be too poor to make it happen.

5.4 The proposition map: when each governance structure is optimal

- **Entrepreneur control** is optimal when private benefits are aligned with efficiency, or when misalignment exists but the investor can still be promised enough expected return.
- **Investor control** is optimal when monetary returns are aligned with efficiency, or when the entrepreneur can still afford to buy back efficient decisions in renegotiation.
- **Contingent control** is optimal in the most interesting middle case: neither side is naturally aligned with total surplus, but the signal is informative enough that authority can be allocated differently across contingencies.

5.5 Financial-contract interpretation

- The paper's applied message is best understood through the mapping from propositions to securities:
  - Proposition 1 $\rightarrow$ non-voting equity,
  - Proposition 3 $\rightarrow$ voting equity,
  - Proposition 5 $\rightarrow$ debt or other contingent-control securities,
  - Proposition 6 $\rightarrow$ debt covenants and charter restrictions in richer environments.
- This is why the paper is so influential: it does not just say "control matters." It says exactly **which financing instruments implement which control regimes**.

6 Short summary

- **Method.** Model a bilateral financing relationship with incomplete contracts, private benefits, zero entrepreneur wealth, and ex post renegotiation.
- **Mechanism.** Control rights determine the status quo in renegotiation, which determines how much surplus can be pledged to the investor ex ante.
- **Main result.** Depending on the alignment of private and monetary returns, the efficient governance structure can be entrepreneur control, investor control, or contingent control. When neither side is naturally aligned with total surplus, contingent control can dominate.

- **Broader implication.** Debt is valuable because it reallocates control after poor performance. More generally, capital structure should be understood as a way of allocating decision rights under contractual incompleteness.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that under incomplete contracts and wealth constraints, **who controls the firm matters for efficiency and for financeability**.
- Cash-flow rights alone do not solve the contracting problem because the entrepreneur values private benefits and cannot always compensate the investor in renegotiation.
- The efficient financial structure is therefore the one that induces the right **governance structure**.

7.2 Economic intuition

- Giving control to the entrepreneur increases flexibility but also increases her ability to extract rents from the investor later.
- Giving control to the investor protects pledgeability but may block efficient decisions because the entrepreneur cannot afford to bribe the investor to change course.
- Contingent control works because it conditions authority on informative performance signals, protecting the investor in bad contingencies while preserving entrepreneurial discretion in good contingencies.

7.3 Takeaway

This paper's lasting lesson is that **capital structure is a control-rights allocation**. The cleanest statement of that idea is the paper's governance pecking order: start with entrepreneur control if that is feasible, move to contingent control if more investor protection is needed, and only then give full control to the investor. In this framework, debt is not mainly a fixed-income claim; it is a device for **transferring control when performance is poor**.